

Attachment 3- Supplemental Contract Information**Contents**

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1.0 TECHNICAL DEFINITIONS and TERMINOLOGY

Acceptable established tree: An existing seedling, sapling or mature tree that is of good form, color, free of damage and has 50% or greater crown ratio. Unless otherwise specified in the Call Order, all visible trees meeting these criteria that are 6 inches and greater will be considered an acceptable established tree.

Access point: The point identified for access to planting areas.

Animal Damage Control (ADC): Methods for protecting the seedlings from animal damage, particularly big game, may include Vexar tubing or flexible netting.

Artificial Shade: Materials not found on the planting site used to shade the tree (e.g. = tree shelters or tree shade cards).

Clean Water and Buckets: Water and buckets used for tree dipping that are free of contaminants that could injure the trees. Water is initially suitable for human consumption.

Clearing: Removal of surface debris around the planting hole to prevent debris from falling into the hole. Debris includes duff, litter, rotten or charred wood, moss, loose rock, ashes, snow, and surface frost.

Container plug: Seedling grown in a receptacle containing soil and nutrients. When the container seedlings are extracted from the receptacle, the soil stays intact around the roots. The soil and roots together are referred to as the container plug; the soil is the container media.

Contracting Officer's Representative (COR): The COR is the on-site contract administrator for the Contracting Officer (CO). The duties and responsibilities of a COR are defined in the letter of designation issued by the Contracting Officer.

Cull Tree: A seedling found at delivery with detrimental molding (mycorrhiza on roots is acceptable), physical damage to the top or roots, excessively dry, of very poor quality, or obviously dead. Cull trees will not be planted or paid for. The Contractor is required to notify and present cull trees found to the Government Inspection Team.

Danger tree: Standing tree that presents a hazard (*OSHA reference 1910.266 C*) to employees due to conditions such as, but not limited to, deterioration or physical damage to the root system, trunk, stem, or limbs and the direction and lean of the tree.

Delivery point: The point identified for delivery of trees and other supplies. If there are multiple points for delivery of various items, it will be so noted.

DBH (dbh): Diameter of a tree at 4.5 feet (breast height) above ground level. On sloping ground, the measure is taken from the uphill side.

Drip line: The line extending vertically from the exterior edge of a tree's live crown to the ground.

Excess trees: Trees that are tallied on the Inspection Plot Sheets that exceed the maximum allowable trees to be planted. These trees are considered wasted trees.

Foreign material (in planting hole): Large rocks, sticks, litter, snow or similar material not part of the surrounding soil substrate that is in the planting hole after the tree is planted.

Ground line of tree before planting: The original ground line of the tree before planting is the soil surface level in the nursery beds or in the container receptacle. It is between the cotyledon scar and the root collar.

Heat Tarp: Tarp with a white upper surface and of silvery metallic reflective underside material designed to prevent solar heating of stored receipted trees. This tarp is designed to be used with the metallic side down over the receipted tree seedlings and the white side exposed to the air. The space blanket will be equal to the Terra Tech "Heat Shield" model.

Inbox temperature: The temperature near the center of the box of trees measured with a probe thermometer or similar instrument. This is the temperature of the seedlings while in the shipping box.

Lateral roots: The horizontal roots extending from the main vertical (tap) root.

Legacy Planting: Technique whereby planting spot selection is based upon favorable microsites (also referred to as “microsite planting”). Planting spot selection will be based upon existing dead, standing, taller stumps, and less mobile downed trees. Utilization of downed material will be used based upon directionality and the ability to provide shade. Logs that lay east/west or northwest/southeast provide the most protection from solar insolation.

M Trees: Abbreviation for measurement of 1000 trees.

Microsite: A plantable spot favorable to seedling survival and growth. Examples of favorable microsites are stumps, down logs, large slash, snags, and standing dead or dying trees that provide protection from sun, wind, animals, and other damaging agents.

Moist soil (in the planting hole): The soil in the planting hole is considered “moist” if it is similar in moisture to the undisturbed soil next to the planting hole at similar depth.

Movable material (also see Transportable Shade): Natural material available on-site which can be moved by the planter. This includes items such as pieces of wood that can be used to provide shade or animal browse protection for a tree. Rock will not be considered movable shade for the purposes of this contract.

Non-planting spot: Campsites, parking areas, fence lines, hiking trails, and roads are designated as non-plantable areas. (See 6.9 for additional non-plantable areas) Where practical, non-planting areas have been deleted from planting unit acreage and may be displayed on the project maps.

Normal planting window: Normal planting window is the estimated number of workdays each season that work can be performed.

Period of Performance (also Performance Period or Contract Time): All calendar days allowed for completion of contract work. Includes all days of the week including weekends and holidays. Calculation allows for workdays, days off, and holidays off.

Plantable spot: An area where a tree can be planted in accordance with all specifications in the contract. In the Planting Inspection method, a plantable spot is also any area where the Contractor has planted a tree.

Planting stock: Government provided seedlings to be planted.

Planting Inspection: The standard planting inspection procedure as described in Forest Service Handbook, R2 Supplement 2409.17 (effective 4/29/2002), 2.82 Part 2. It includes the description of planting violations that result in unsatisfactorily planted trees, plot sizes, calculations for determining planting quality based on trees planted, and similar information. Example of inspection form R4-FS-2470-9 (6/2001) is included in Attachment 2, Appendix A of this contract.

Protection: Natural and artificial materials that act as an obstacle protecting the tree from animal or other injurious agents. The protection material and effort required by the planter to protect the tree varies depending on the site conditions of each unit.

Quality Assurance: The actions taken by the Government to assess the Contractor’s results to determine that they meet contract requirements. The methods for quality assurance are described in the Quality Assurance Surveillance Plan (QASP).

Quality Control: Those actions taken by a Contractor to control the production of outputs to ensure that they conform to the contract requirements. The methods for inspecting for quality control are described in the Contractor’s Quality Control Plan (QCP).

Range of Work Season: The range of the work season is the estimated month and day of the earliest beginning and latest ending when contract work could be performed.

Residual trees: Live trees remaining in the unit after the harvest or fire activity.

Root collar: The transition zone between stem and root. In tree seedlings it is identified by the presence of a slight swelling.

Root length: The root length measured from the original ground line to the end of the main vertical or tap root.

Satisfactorily planted tree: A tree seedling that is planted in full accordance with all specifications as set forth herein.

Scalping: The removal of live vegetation, including roots, to reduce competition from existing vegetation and to expose mineral soil.

Shade: Natural and artificial materials that protect the tree from the sun's heat and wind. The type of shade material and effort required by the planter to shade the tree varies depending on the site conditions of each unit.

Soil Surface: Where surface debris is present, the soil surface is that point where leaves, needles, twigs, wood, and other surface litter has decomposed to the point that the original litter material cannot be recognized with the naked eye.

Stationary material: Natural material used to shade or protect the tree that is not moveable by the planter. Natural material includes items such as dead standing trees, down logs, dead brush, or terrain features used to provide the required protection for a tree. Stationary material generally offers the most shade and protection for seedlings from sun and animal trampling.

Top height: The length from the original ground line to the top of the terminal leader; does not include needles extending above the leader.

Transportable shade: Natural material existing on-site such as wood debris that can be moved to the planting spot to shade the tree from the sun. Also known as "moveable shade".

Treatable Seedling – A healthy and vigorous seedling that is a minimum of 4 inches tall. Seedlings with minor preexisting browse damage may be considered treatable if there are no acceptable undamaged seedlings nearby that could be selected instead.

Trees in possession of the Contractor: The time after trees has been issued to the Contractor. During this time, the Contractor is responsible for all aspects of tree care and accountable for tree quantities.

Under-planting of seedlings: Planting underneath a stand of trees. Planted trees must be placed in a mineral soil area, which may require the clearing of some vegetation or slash with the planting tool. Call Orders will specify distance from dripline of overstory trees.

Unplantable spot: An area within the specified spacing limit in which it is not possible to plant a seedling according to specification and no tree has been planted. One unplantable spot is allowed for each single unplantable area equal in size to the average specified spacing. Example: if a single unplantable area of 64 feet exists on an 8-by 8 foot spacing, one unplantable spot is recognized.

In hoedad planting, a spot is considered unplantable if a suitable spot for planting a tree cannot be found after three attempts of throwing the hoe, and if the hole cannot be opened at the spot with five swings or less.

In auger planting, the spot is considered unplantable if a suitable plantable spot cannot be found after three attempts at clearing or scalping, and then after three attempts to open the hole in the spot with the auger fail.

Unsatisfactory planted tree: A planted tree which fails to meet one or more of the specifications of a satisfactorily planted tree. Refer to definitions for planting violations described by Planting Inspection in FSH 2409.17, 2.82 (R2 supplement).

Wasted material: Material provided to the Contractor to conduct a task such as installing artificial shade that is lost, dropped, or damaged by the Contractor or which is installed on trees not in accordance with the requirements of the contract. This may include, but is not limited to, installing shade on the wrong tree species.

Wasted tree: A tree that is lost, damaged, destroyed, or handled contrary to the standards for care of trees. Planted trees that exceed the maximum number of trees in the Planting Inspection plot procedure are also considered wasted trees.

2.0 CONTRACTOR FURNISHED PROPERTY

The Contractor furnished tools and equipment will be specified in the Call Orders. This may include but is not limited to:

Item	Description
Hoedads/ Planting bars	The Contractor shall provide hoe-dads or planting bars that have a blade with a minimum 3-inch width and a minimum length of 2 inches longer than the container plug to be planted.
Power Augers	If Power Augers are used, specifications will be provided in Call Orders. If used, they shall be lubricated with vegetable oils or similar substances not injurious to trees and shall not leak gas.
Dipping tubs, wetting agent and clean water (Bareroot only)	The Contractor shall provide dipping tubs (plastic cans). Dipping tubs shall have a minimum capacity of 76 liters (20 gallons) and shall be free of chemicals and petroleum products. Contractor shall furnish wetting agent (vermiculite) for dipping trees in. Vermiculite shall be of fine or medium grade. Water should be clean and suitable for human consumption. The Call Order shall specify whether to have the contractor dip trees or have them planted directly from the shipping container and what the ratio of water

Item	Description
	to vermiculite will be.
Planting bags	The Contractor shall provide planting bags. Planting bags shall be a light color and shall have a minimum depth of 38 cm (15 inches) and shall be free of excessive wear. All planting bags shall have polyurethane foam liners that line the interior sides and bottom of bags, or other insulated liners acceptable to the Contracting Officer. Burlap is not acceptable. Planting bags and liners shall be clean and free from dirt and residue from prior planting operations when the call order begins, or after any stoppage of work where the Contractor plants elsewhere.
Transportation	The Contractor may need to provide transportation from the delivery point(s) (insulated van or enclosed trailer, free of oil and/or gas residue and/or fumes) that can transport a minimum of 30 boxes of 1-0 plugs) for trees seedlings and other government furnished property to access point(s). The Call Order shall specify whether Government or Contractor transport will be used.
Heat Tarp	The contractor shall provide 2 heat tarps per crew to use to protect receipted planting stock stored in the open prior to seedlings being planted.
Miscellaneous	The Contractor shall provide drinking water, food, transportation and housing for workers.

3.0 GOVERNMENT FURNISHED PROPERTY and SERVICES

The property and services that will be provided by the Government will be specified in the Call Order and may include but will not be limited to:

Item	Description	Where / When
Detailed Information Chart	Chart with unit specifics.	Provided within Call Order
Project Maps	Maps of planting units with access roads.	Provided in Call Order. Additional maps may be provided prior to planting.
Access Road(s)/ Access Point(s)	Access roads may become impassable when wet, snow covered, or when wind thrown trees fall across the road. The Government will initially clear down trees from access roads or plow to provide access to planting areas. After initial Government clearing of access roads, the Contractor shall clear any subsequent falling of trees that block them. Access roads are accessible by 4-wheel drive vehicles equipped for mountain driving. Snow tires and chains may be needed in inclement weather.	Prior to and during range of work season.
Access behind locked gates	For units behind Forest road closures (locked gates) The Government will provide access through locked gates unless walk-in is required.	During period unit is being planted.
Bareroot grown tree planting stock	<u>Bareroot grown tree</u> : Size of trees will be within the range specified in the <i>Schedule of Items</i> . Ninety percent of the trees will have tops ranging from 4 to 16 inches Ninety percent of the trees will have vertical and lateral root lengths from 8 to 12 inches. Occasional lateral roots on individual trees may exceed this length. When an individual tree is encountered with a vertical root or a majority of the lateral roots exceeding 12 inches it shall be planted in accordance with clause 6.4.3 or it may be returned to the Government.	Delivered to delivery/access points except as noted in 4.1.4 (below).
Container grown tree planting stock	<u>Container grown tree</u> : Size of containers will be within the range specified in the <i>Schedule of Items</i> . Depending on container size, trees may or may not be delivered to Contractor extracted from the containers. Generally, the small containers will be extracted, while larger ones may not.	Delivered to delivery/access points except as noted in 4.1.4 (below).
Tree wrapping materials (Bareroot only)	The Government will prepare the trees by dipping them in a wetting agent (vermiculite, terra sorb) and wrapping (kintex) them. Each piece of wrapping material shall be returned to the delivery/access points after the trees from that bundle have been planted.	Delivered to delivery/access points except as noted in 4.1.4 (below).
Artificial Shade or Browse Protection Material, if required.	Shade tubes, Browse Protection (both with stakes), Shade Cards with wire wickets. Other tools might include staple pliers, staples, or compasses.	Delivered to delivery/access points except as noted in 4.1.5 (below).

Item	Description	Where / When
Planting Inspection Sheet (Attachment 2, Appendix A)	Planting Inspection Sheet. Quantity necessary to inspect the contract.	At pre-work meeting.
Artificial Shade or Protection Inspection Sheet, if required.	Shade or Protection Inspection form. Quantity necessary to inspect the contract.	At pre-work meeting

4.0 CONTRACTOR WORK REQUIREMENTS AND STANDARDS

4.1 GENERAL

4.1.1 Road Access. The Contractor may access the units from forest roads indicated on project maps. Some units may require the Contractor to walk-in, if specified in the Call Order; access is to the access point noted on project map.

Tree transport within units: The Contractor may transport trees from the access point(s) to portions of the unit at the Contractor's discretion consistent with any restrictions on motorized or other vehicle type specified in the contract/Call Order and with COR approval. If stored for more than 15 minutes, the trees shall be stored in the shade or covered by a heat tarp and shaded from solar insolation.

4.1.2 Walk-in units. These units cannot be accessed from forest roads. The Contractor shall transport trees from the delivery points to access points using the method of his/her choice consistent with restrictions on motorized vehicles or other transport method noted on the Call Order. When applicable, the COR may authorize the use of ATVs to support transport of the seedlings into these units or the Government may choose to use ATVs to transport the seedlings to the unit, Government provided pack strings may be available for use to transport trees from access points to planting units. These conditions will be described in the *Detailed Information Chart*. The Chart provides the distance in feet from a Delivery Point based on path of least resistance. The Chart also indicates alternative walk-in distances if snow cover prevents vehicular access to the Delivery Point of the sub-item.

4.1.3 Locked gates. In cases where units are behind locked gates, the Contractor shall close and lock gates after entering or exiting each gate and will be subject to all other restrictions of administrative use. Unless otherwise stated in the Call Order, the Contractor will be permitted to drive from the locked gate to the unit access point with only the minimum number of vehicles necessary for efficient transport of the crew. No firearms are permitted behind locked gates. Method of providing access behind locked gates will be identified in individual Call Orders or discussed at prework meetings.

4.1.4 TREE ORDERS AND DELIVERY. The Contractor shall order trees at least **12 hours** in advance of desired delivery. This time period is necessary to allow the Government time to assure trees are prepared for planting. The Contractor will identify which planting units are to be planted with the trees ordered such that the Government can deliver trees of the proper seed lot for those planting units.

- (a) If specified in the Call Order, the Contractor shall transport planting stock from the delivery points to access points in the manner agreed to with the Contracting Officer. Planting stock shall be cared for and protected as specified in **section 4.2 (below)**. The Contractor may pick up planting stock for the current planting day during a time period specified in the Call Order or as otherwise agreed to with the COR. The Call Order shall specify which method will be used (government transport or contractor transport).
- (b) The Contractor shall plant all planting stock ordered and delivered to the delivery/access points within **12** hours of delivery, unless otherwise approved by the Contracting Officer or COR. Seedlings shall not be returned to the Government for cooler storage unless approved by the COR. When the COR or Contracting Officer determines that conditions are unsuitable for planting, the Contractor shall return all unplanted planting stock to the delivery points with no penalty for delay in planting the delivered planting stock.
- (c) If Call Orders specify Government transport of trees, the Government will deliver trees to access points identified on project maps.

4.1.5 Artificial shade or protection materials orders and delivery. The Contractor shall request materials for installing artificial shade or protection materials at least **12 hours in advance** of desired delivery. Material will be delivered to the access points indicated on the project map.

4.1.6 Return of Government furnished property. The Contractor shall return unused Government supplies, unplanted trees, all packaging materials, and wrapping materials to the access point or COR at the end of each planting day. If applicable, gate locks and/or keys shall be returned after completion of the planting unit accessed by the gated road.

4.2 TREE HANDLING AND CARE

4.2.1 The Contractor shall be responsible for the care of trees from the delivery points and throughout the time that the stock is in possession of the Contractor to assure they are properly cared for and not damaged or exposed to damaging conditions consistent with Forest Service Handbook (FSH) 2409.17, 2.53 (R1 supplement). Trees not properly cared for may be considered wasted.

4.2.2 Trees shall not be altered (includes root or top pruning). If pruning or culling appears necessary, or if mold, dry roots, evidence of injury, or drying is seen, the condition shall be reported to the COR.

4.2.3 Contractor shall take precautions to keep roots moist during the entire planting operation.

4.2.4 Planting stock shall be protected from drying, heating, smothering, freezing, crushing, bending, drowning, abrasion, rapid temperature fluctuations, or contact with injurious substances.

4.2.5 Trees that are frozen shall not be planted. They shall be thawed in accordance with COR direction.

4.2.6 Shipping containers containing planting stock shall be opened in full shade. Protect containers from direct sunlight, and from being tossed or dropped. Containers shall be promptly resealed after removing plants.

4.2.7 Outside storage units, the temperature in the shipping boxes shall be maintained below 45 degrees F; or similar to the temperature in the box at receipt, whichever is higher.

4.2.8 Wrapped or bagged planting stock will have the wrapping loosened just prior to planting to allow plants to be easily extracted without damaging the soil plug or roots; however, planting stock shall remain wrapped or bagged while in the planting bags.

4.2.9 Larger container planting stock that are not extracted may have to be carried to planting hole in containers and extracted at the hole. Contractor shall carry empty containers back to the access point as designated in the Call Order.

4.2.10 Trees carried in bags shall not exceed the amount that can be carried and removed without injury, nor exceed the amount that can be planted before critical heating or drying occurs. Planting stock in planting bags shall only have the tops exposed.

4.2.11 Planting stock shall not be removed from the planting bag until immediately before planting in the prepared hole. Planting stock shall be handled carefully to avoid damage to roots and buds. Trees with mold or evidence of injury shall be reported to the COR.

4.2.12 Trees placed in planting bags will be planted and not returned to storage except as approved by the COR or Contracting Officer. When returning planting stock is allowed, they shall be returned to shipping boxes marked to indicate seed lot or other information specified by the COR or Contracting Officer.

4.3 TREE PREPARATION

4.3.1 Water dip at planting site. The Contractor shall dip the roots of bare-root trees in clean water or clean water with wetting agent at the planting site, prior to placing trees in the planter's bag. Dipping in water and/or with wetting agent is intended to increase root moisture. Roots shall be fully wet, but soil particles shall not be washed off the roots.

(a) Contractor shall provide clean water and dipping bucket(s) sufficient for dipping trees.

4.4 CLEARING

Before preparing the planting hole, all debris down to the soil surface in a spot equal to or larger than minimum scalp size specified on the Call Order, ***Detailed Information Chart*** shall be removed. Debris to be removed includes duff, litter, rotten or charred wood, moss, loose rock, ashes, snow, and surface frost. If the clearing of branches and limbs can be done with reasonable effort, without setting aside the scalping tool, the spot shall be considered plantable. The distances may differ from those stated in FSH 2409.17, 2.62 (R2 supplement) for both clearing and scalping. The Detailed Information Chart rules.

4.5 SCALPING

Each planting spot shall be free of live vegetation, including roots, to reduce competition from existing vegetation. Mineral soil shall be exposed over the scalped area. Trees may be located toward the edge of the scalp to meet Stationary shade requirements. Where Transportable or Artificial shade is used, the seedling should be planted in the center of the minimum scalp. Unless otherwise specified in the Call Order ***Detailed Information Chart***, the minimum diameter of the scalp will be 16 inches with a maximum depth as needed to expose mineral soil and competing surface vegetation.

4.6 TREE PLANTING

To give the best chance for tree survival and growth, trees shall be planted according to the following standards:

4.6.1 Satisfactorily Planted Tree. Trees shall be planted satisfactorily both above and below ground (Attachment 2, Appendix D) or as defined in **FSH 2409.17, 2.67 (R2 supplement)**. Trees shall be planted in moist soil, free of foreign debris. Trees shall be planted, in an open hole, deep and wide enough to fully accommodate the roots of the tree being planted. Tree roots shall be in their natural configuration and not wedged or jammed into the side of the hole and shall meet all requirements of the contract. The open-hole method shall be used as described in FSH 2409.17, 2.63 (R2 supplement). Planting hoes shall have a 4-inch-wide planting blade that is 2 inches longer than the container plug in length. Power augers shall be capable of drilling a 4-inch diameter hole to a depth 2 inches deeper than the specified root length. Augers shall be lubricated with vegetable oils or similar substances not injurious to trees and shall not leak gas.

4.6.2 The hole shall be filled and firmed with moist soil, so the tree is firmly in the soil and there are no air pockets. There shall be no foreign material in the hole such as dry soil, ash, organic matter, snow, and/or rock.

4.6.3 Bare-root trees shall be planted deep enough so that after filling and firming, the soil surface is at or slightly above the original ground line of the tree (Attachment 2, Appendix D). No portion of the roots shall be exposed, nor any branches covered with firmed soil. The roots shall not be doubled up, twisted, or spiraled. The root system shall be aligned with the axis of the planting hole with all roots extending downward. When trees are planted with lateral roots exceeding the maximum length specified in clause 5.0, the long lateral roots shall be placed in the bottom of the planting hole in a manner that does not affect the placement of the vertical root or other lateral roots. Trees with long lateral roots planted in this manner will not be subjected to a root configuration violation during inspection.

4.6.4 Container trees shall be planted so that after filling and firming, there is ½ inch to 1 inch of soil on top of the container plug. The Contractor shall take care to keep the container growing media intact but where the media falls away from the roots, the tree shall be planted like a bare-root seedling with ½ inch to 1 inch deeper than the original ground line (Attachment 2, Appendix D).

4.6.5 After planting, the tree stem shall be erect and oriented between vertical and perpendicular to the slope. The tree shall not be weighted down with mud or debris. There shall be no excessive depressions on the ground surface that may collapse on the tree hole and the tree shall not be in contact with the dirt bank. The tree shall be free from damage.

4.6.6 All additional or auxiliary holes created during the planting operation that are within 18 inches of the planted tree, shall be filled-in or collapsed.

4.7 ON-SITE SHADE REQUIREMENTS

4.7.1 Unless otherwise specified in the *Detailed Information Chart*, microsite planting and shade is very important for bareroot and containerized seedlings in most situations on the Forests due to high elevations, spring winds, and for animal protection. Trees shall be planted on the north, northeast, or east side of the shade material. Each tree shall be located as close to the shade material as possible, and a **maximum** of 2 inches away from shade material. Use of shade will be emphasized in most units/sub-items.

4.7.2 Description of Shade Requirements. Trees shall be planted consistent with the shade and protection requirement for each planting unit as specified in the Detailed Information Chart. The type of shade material and effort required by the Planter to shade the tree varies depending on the site conditions of each unit (Attachment 2, Appendix E). Natural shade material includes downed logs, large slash, stumps, standing snags, shrubs and rock meeting minimum specifications. Unless otherwise specified in the Call Order, trees will be planted in the following conditions. Natural microsite shade and protection material includes:

- Down logs with a minimum small-end diameter of 6 inches and minimum length of two feet.
- Large slash in contact with the ground surface, small-end diameter of 4 inches and minimum length of two feet.
- Stumps with a minimum diameter of eight inches and minimum height of eight inches.
- Standing dead trees at least eight inches in diameter at breast height.
- Shrubs and rock may be used as a tertiary shade item if specified in the call order.
- When shade is required, each tree shall be shaded at the ground level from the afternoon sun. Trees shall be on the N, NE, or E side of the shade material. If multiple stationary material is available, the planter shall plant the seedling next to the largest diameter material present. Each tree shall be as close to the shade material as possible, and the maximum within ¼ of the distance equal to the height of the shade material (for stumps, downed logs and similar) or ½ the distance equal to the width of the shade material in the case of standing snags. Seedlings planted next to down logs shall be spaced from those seedlings planted next to the same log the standard spacing distance.

- When transportable (shade) material is required, it must be a piece of wood at **least 4 inches in diameter**; if multiple woody pieces are available within the 10-foot radius, the largest movable piece must be used. The Contractor shall move suitable pieces of wood that can be carried or rolled to the planting spot, provided it is within a **10-foot radius** of the planting spot and is not providing protection to another planted tree. All transportable shade material shall be of such size and/or arrangement so as to be reasonable sturdy and not easily toppled or displaced by animals, wind and/or snow. The shade material shall be arranged with consideration of planting spot topography so as to prevent it from rolling or falling on top of planted tree seedlings or from rolling or falling out of shading distance from the seedling.

4.7.3 **Planting Spot Selection Priority.** Natural Shade and Protection Requirements.

1. Stationary material first, then transportable shade. A tree shall be planted next to stationary material when it is available. When stationary material does not exist in the spacing allowance, shade shall be transported to the seedlings. When neither stationary nor transportable material exists, no tree shall be planted.
2. Stationary material first, transportable shade second, plant in open last. A tree shall be planted next to stationary material when it is available. If stationary material does not exist in the spacing allowance, shade shall be transported to the tree. When neither stationary nor transportable material exists, the tree may be planted in the open.
3. Stationary material only. Tree shall be planted next to stationary material when it is available. When stationary material is not available, no tree planted unless specified in the Call Order.
4. No shade. There is no requirement to select the planting spot based on shade or protection.
 - If Stationary material or transportable material is unavailable, the tree may be planted in the open, unless specified in the Call Order and/or Artificial Shade Installation is required.

4.8 SPACING GUIDELINES

Where per acre planting is performed, trees shall be planted to achieve the average spacing specified in the **Detailed Information Chart** in the Call Order. It is expected that an irregular distribution will result as planters avoid acceptable established trees, rock or shallow soils and seek planting spots with the required microsite protection materials as well as planting spots in the best microsites and soil conditions. Spacing for individual trees may vary as much as 50 percent in any direction to attain the best plantable spot. The number of trees planted shall not be materially changed by the method of selecting planting spots. Each call order shall identify the spacing and stocking requirements.

Where per tree planting is performed, trees shall be legacy planted using On-Site Shade Protection – Planting Spot Selection Priority 2 as described in section 4.7.3 or as described in each call order. Spacing will be highly variable and may be dependent on aspect. The west to northeast aspects may naturally have a higher availability of planting spots than east to southwest aspects. Planting spot selection takes precedence on location of planted trees.

4.9 NON-PLANTING AREAS

Within certain planting units, areas of no tree planting are identified due to existing advanced regeneration, dwarf mistletoe infestations, wet areas, loose sod rolls, deep woody debris beds, unburned slash piles, burn pile ash pits, untreated roadbeds, and ephemeral draws. Where practical, non-planting areas have been deleted from planting unit acreage and may be displayed on the project maps.

4.9.1 Advanced Regeneration. Where extensive areas of acceptable seedlings and saplings already exist inside a planting unit, these areas are identified for non-planting. The Planters will walk through checking for open areas which can be planted. See Call Order for planting distance from live conifer and aspen trees.

4.9.2 Non-planting Areas Adjacent to Dwarf Mistletoe Infested Stands. As shown on the project maps and described in the Detailed Information Chart, no trees will be planted within 10 to 100-feet (depending on the specific planting unit) of the edge of the planting unit boundary. These no planting buffers have been identified by flagging on the ground.

4.9.3 Glacial Potholes and Wet Areas. No seedlings are to be planted below the high watermark as evidenced by standing water or a change to wet soil type vegetation. Identified potholes are shown on the project maps. Where drainage water is found running downhill of road culverts, these wet areas are considered non-plantable.

4.9.4 Sod Rolls in Site Prepared and Recent Timber Harvest Units. The sod rolls created during dozer site preparation for planting and from recent timber harvesting are non-plantable. Seedlings will not be planted on the sod rolls.

4.9.5 Deep Woody Debris Beds. Examples of fine woody debris include beds of saw dust, wood chips, shaved off bark, fine twigs and branches. These woody debris beds will be encountered in tree processing and log landing areas. If the Planter is unable to clear woody debris to a suitable planting spot in three attempts with a planting tool, then the planting spot is non-plantable.

4.9.6 Unburned Logging Slash Piles. Areas within 30 feet of unburned logging slash piles will not be planted since the seedlings maybe damaged by heat during pile burning.

4.9.7 Burn Pile Ash Pits. The ash pits left after slash pile burning are considered to be non-plantable due to poor survival of seedlings. The lightly burned shoulders are plantable if the spot can be cleared. Where ash pits have been ripped and stirred by heavy equipment prior to planting, the ash pit site is considered plantable.

4.9.8 Untreated Road Beds. Roads are considered non-plantable if they are open to motorized travel, or where closed if the roadbed has not been ripped to reduce soil compaction.

4.9.9 Ephemeral draws. Sandy washes of intermittent and ephemeral draws, especially of decomposed granite are non-plantable areas, seedlings can be planted within 1 foot of the edges of these washes.

4.10 ACCEPTABLE ESTABLISHED TREES/OTHER PLANTED TREES

Seedlings will not be planted under the drip line of larger, acceptable, live conifer residual trees. An acceptable established tree is a conifer seedling or sapling that is of good form, color, free of damage, has 50% greater crown ratio, and is the species and size specified in the *Detailed Information Chart*. Unless otherwise specified in the Call Order, seedlings will not be planted within 100 feet of acceptable aspen or other acceptable hardwoods identified in Call Orders. An established tree is treated as an unplantable spot, unless specified in the Call Order.

4.11 SPECIES OR STOCK TYPE MIX

The species and stock type to be planted is indicated on the *Detailed Information Chart*. Where a mixture of species and/or stock type is required, the Contractor shall distribute the variety of species and/or stock type over the entire unit except when a geographic location is specified in the *Detailed Information Chart*. When a location is noted, the Contractor shall plant according to the specifications.

4.12 TREES NOT MEETING SPECIFICATIONS

The Contractor is not obligated to plant trees that do not meet the specifications and they shall be returned to the Government. If trees not meeting specifications are planted, they shall be planted satisfactorily.

4.13 ARTIFICIAL SHADE OR PROTECTION INSTALLATION

If required in the Call Order, artificial shade shall be installed only on trees not protected by existing on-site stationary or acceptable transported material.

4.13.1 Tree shade cards shall be installed concurrent with planting to protect trees from the afternoon sun, between approximately 2 - 4 pm. The cards shall be within 2 to 4-inches of the tree and positioned to shade the tree from the afternoon sun. Placement will be south-southwest of the seedling on azimuth $210^{\circ} \pm 30^{\circ}$. Wire wickets will be set deep enough into the soil that the base of the mesh shade card touches the soil surface. The plastic mesh of the tree shade card will be stapled at least twice over the top of the wire wicket to prevent the mesh from being easily pulled by wildlife and livestock. Tree shade cards shall be installed so as not to damage the tree including skinning the bark, breaking the leader, damaging roots, or bud stripping. Wickets holding the card shall be **3-4** inches away from the tree to avoid root damage. Do not open the mesh shade card and cover the seedling inside. In the event that other methods of artificial shade not described above will be used, a description and installation specifications will be included in the call order.

4.13.2 Tubes shall be installed so as not to damage the tree including skinning the bark, breaking the leader, damaging the root, or excessive bud stripping. The terminal buds shall be below the top of the protector to protect against browse. Terminal buds and all lateral branches shall be upright inside the protector and shall not be bent over, twisted, or otherwise damaged. The tubes shall be held in place with one (1) Cedar Arrow Shaft or 1 Bamboo Shaft. The shaft shall be woven in and out of the tube a minimum of four (4) times. The stake holding the tube shall be **3-4** inches away from the tree to avoid root damage. The shaft shall be driven into the ground six to nine (6 to 9) inches. The completed tube and support shall be in a near vertical position when done. Spacing can be varied up to 25% to select the best treatable seedlings, while still evenly spacing the protectors throughout the unit.

4.14 REMOVAL OF ARTIFICIAL SHADE OR PROTECTION INSTALLATION

4.14.1 Shade Card Removal. Remove all shade cards and stakes without damage to the tree or root system. Shade cards must be removed from site in bags provided by the Contractor and delivered to a designated drop point. Other removal criteria and drop point location(s) will be specified in the Call Order.

4.14.2 Tube Removal. Remove all tubes, plastic ties and stakes without damage to the tree or root system. Wood stakes may be left on site. Tree shelters and ties must be removed from site in bags provided by the Contractor and delivered to a designated drop point. Other removal criteria and access point location(s) will be specified in the Call Order.

4.15 WORK PLAN AND SCHEDULE

The Contractor shall prepare a work plan and schedule that demonstrates that work will be completed within the performance period. The Contractor shall provide a general plan with the technical proposal. A supplementary plan and schedule shall be provided after issuance of contract award, but within two days of beginning work.

4.16 CAMPING AND WORKSITE CONDITIONS

The Contractor shall prepare a Camping and/or Worksite Plan and operate in a manner that will protect the environmental resources. In work camps, portable toilet facilities must be provided and maintained by the Contractor. Pit toilets or free range around camp is not acceptable. Portable toilets-on-trailers are available for rent from local sewer and septic businesses in the communities. Due to black bear activity, food and refuse will be secured in a protected location. On Forests with grizzly bears, additional food storage restrictions may be required in the call order.

4.17 COMMUNICATIONS

The Contractor shall keep the Contracting Officer advised as to where the Contractor can be reached prior to start of Contract Time. Further the Contractor shall maintain contact with the Contracting Officer during periods of work suspension to reduce any delay in resuming planting and to determine the best suitable time and conditions to resume planting.

4.18 SUPERVISION AND ON-SITE REPRESENTATIVE

At all times during performance of this contract and until work is completed and accepted, the Contractor shall directly supervise the work or assign and have on the worksite a competent superintendent who is satisfactory to the Contracting Officer and has authority to act for the Contractor. The Contractor's on-site representative and inspector shall be fully conversant and literate in the English language and shall be in the work area whenever work is in progress.

4.18.1 All work will be performed in an organized and systematic manner. Planters will generally all be working in the same unit except when authorized by the Contracting Officer or their designated Representative.

4.19 CREW SAFETY

The Contractor is at all times responsible for the safety of their crew and will consider and address crew safety and provide a Safety Plan recognizing the hazards as stated in each Call Order.

5.0 CALL ORDERS

The Contracting Officer shall submit an executed call order to the Contractor. All work will be ordered by Firm-Fixed price at the prices listed on the Schedule of Items. The price may change only by a written modification to the call order signed by the Contracting Officer and the Contractor.

6.0 CONTRACTOR QUALITY CONTROL

The Contractor shall provide quality control for all planting operations including tree care, handling and planting. The Contractor is responsible for the quality of planting on the site and shall conduct prompt plot inspections to ensure proper planting procedures are being followed.

6.1 QUALITY CONTROL PLAN

The Contractor shall provide a general Quality Control Plan (QCP) with the technical proposal. The Contractor shall submit the QCP within **20 days** after contract award based on items awarded to the Contractor and Government comments. The general Quality Control Plan must demonstrate how the Contractor will assure quality meets the requirements and specifications of the contract.

6.1.1 Plot Inspections Tree Planting. The Contractor shall maintain a plot system that provides an unbiased sample of planting quality, and which represents the entire area planted. The plot system shall be consistent with the planting inspection method, specified in FSH 2409.17, 2.82, part 2 (R2 supplement, found in Attachment 2, Appendix B). Contractor will sample **at least 1-percent** of each planted sub-item, with a minimum of 5 plots per sub-item. The plot size will be specified in the Call Order. The Government may observe or re-examine the Contractor's plots at any time. The Government may combine sub-items, due to small acreage size (>9 acres), unless otherwise specified in the Call Order, each sub-item/unit will be assessed separately, and results will not be averaged with those of other subitems:

- (a) Plots shall be marked and identifiable on the ground.
- (b) The Contractor will provide written inspection data by unit/sub-item using the *Planting Inspection Sheet* (R1-2470-9), Attachment 2, Appendix A, and will include a sketch map with plot locations to the Government by 8:00 a.m. the day after inspection is completed on the sub-item/unit. Inspection sheets shall be signed and dated by the person who conducted the inspection and shall certify that the inspection records are complete and accurate.
- (c) Calculation of planting quality will be based on the following formula (refer to columns of *Planting Inspection Sheet*). All computations will be carried out to at least thousandths (.000) and the final percentage of planting quality will be rounded to the nearest whole percent.

No. of Satisfactory Trees Above Ground (Column 9)	X	No. of Satisfactory Trees Below Ground (Column 11)	X 100 = Quality %
Actual No. of Plantable Spots on which Trees should have been Planted (Col. 5)		No. of Trees Dug (Column 10)	

- (d) Plot inspection will include a tally of trees planted that are in excess of the maximum allowable trees to be planted on each plot and noted in Column 8 of the Inspection Sheet. These excess trees are considered wasted trees. To ensure that trees are distributed throughout the unit fairly consistently, wasted trees shall be tallied by plot. The sum of wasted trees on all plots on the unit/ sub-item will be used to determine the damage assessment.

6.1.2 Inspections- Artificial Shade/Protection Installation. In those sub-items where Artificial Shade/Protection installation is required, inspection of the installation will be concurrent with the tree planting inspection. Each tree within the tree planting inspection plot will be inspected for acceptable Artificial Shade/Protection installation.

- (a) Plots shall be marked and identifiable on the ground.

- (b) Artificial Shade will be inspected for assembly including stapling, positioning (azimuth and distance), firmness, and depth. Failure to properly install the Artificial Shade/Protection will be recorded as a “Shade/Protection Violation” “X” on the Planting Inspection Sheet (R1-2470-9), Exhibit 01.
- (c) Protection Installation will be inspected for assembly including: Cedar shaft firmly in the ground (6-9 inches). Protector is securely woven to the cedar shaft. Protector and shaft are near vertical (within 5 degrees). The seedling has not been damaged during installation of the protector. Terminal bud and lateral branches are upright and are not entangled in the protector. Protectors are evenly spaced throughout the work unit. Protector is installed on a treatable tree. See definitions.
- (d) Calculation for percent quality for installation of artificial shade/protection material will be based on the following formula.

$\frac{\text{Total No. of Satisfactorily Shaded or Tuber Trees}}{\text{Total No. of Trees Required to be Shaded or Tuber}} \times 100 = \text{Quality \%}$
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- (e) The inspection shall include a tally of wasted materials due to: a) excess or unspecified material used or treated within the inspection plot, and b) materials dropped or damaged by Contractor. Wasted material shall be tallied by plot. The sum of all plots on the unit/ sub-item will be used to determine the damage assessment.

6.1.3 Acceptance. See Quality Assurance Surveillance Plan for information on acceptance.

6.2 REWORK OF UNACCEPTABLE WORK

See Quality Assurance Surveillance Plan for information on rework.

7.0 MEASUREMENT

See Quality Assurance Surveillance Plan for information on measurements used in the contract.

7.1 RE-MEASUREMENT

See Quality Assurance Surveillance Plan for information on re-measurement of units.

10.0 METHOD OF CALCULATING PAYMENT

See Quality Assurance Surveillance Plan for information on payment calculations.

10.1 TREE SHELTER INSTALLATION / REMOVAL – PAYMENT CALCULATION

This item is not included in this contract.

10.2 UNACCEPTABLE WORK

See Quality Assurance Surveillance Plan for information on how unacceptable work will be handled in the contract.

11.0 DAMAGES

See Quality Assurance Surveillance Plan for information on how damages will be handled in the contract.

